Iris_to_CAT

March 20, 2024

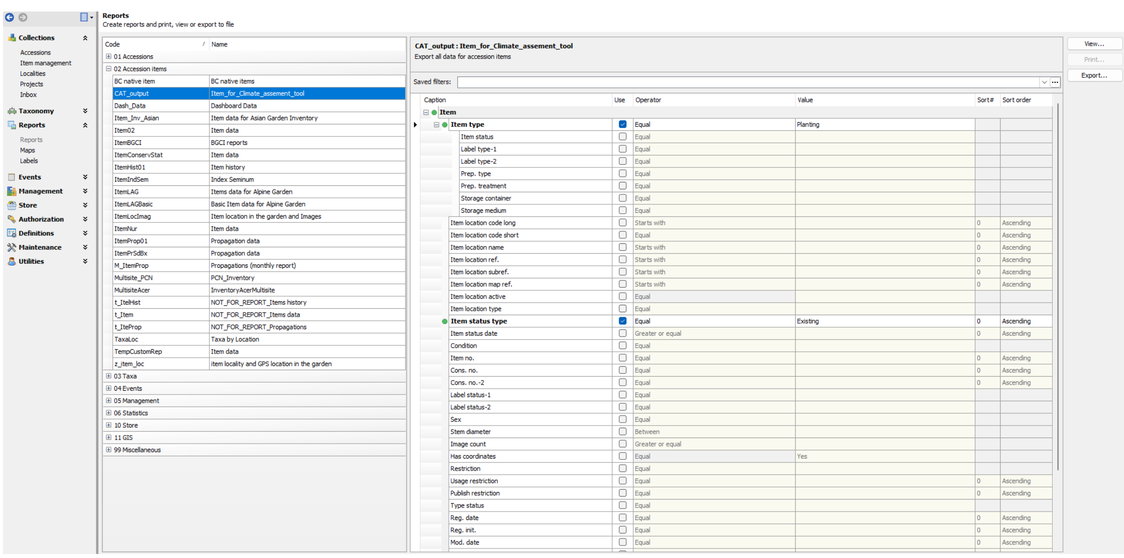
1 How to Use the Climate Action Tool from IrisBG to CAT

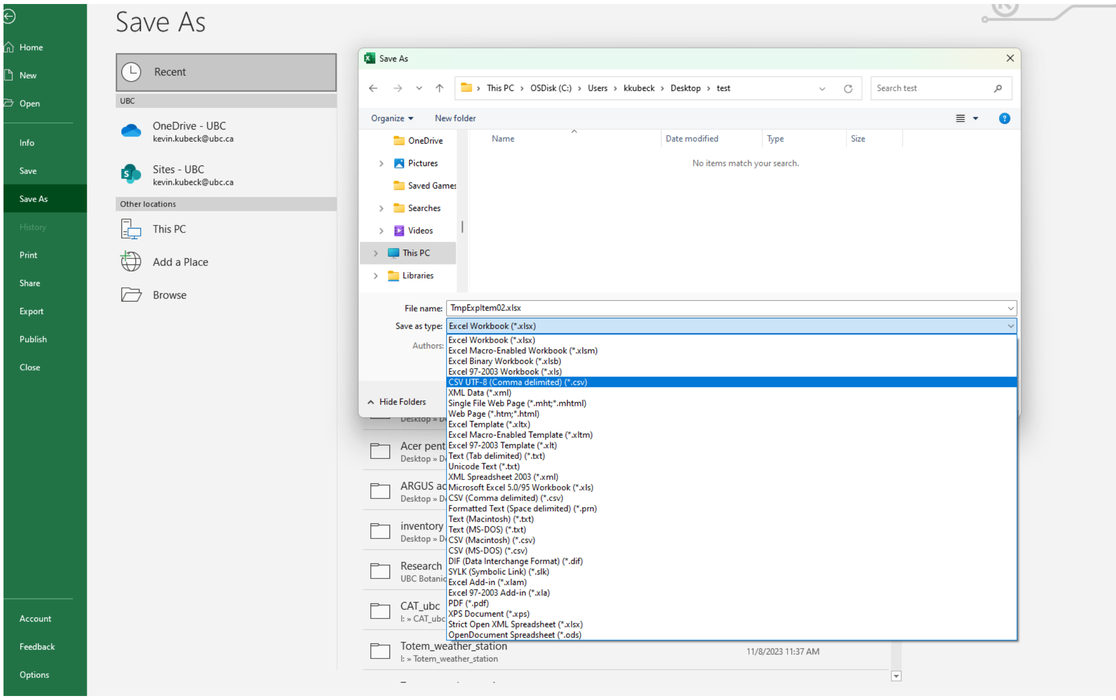
1.1 The Data Creating and Pre-processing

Use the report system from IrisBG to create an Excel or CSV file. The example shown uses the Python library Pandas for data manipulation but similar approaches can be made using Excel.

Use an Item-based search if you are looking for actual plants in the garden, historical or current. - Use lifeform data field to filter tree, shrubs etc. Only woody forms are currently in CAT. - Remove multiples of the same taxon to avoid repeated queries - Simplify taxonomy. CAT is expecting a Genus and a species

Use a Taxon-based report for a more generalized approach. - Lifeform data field to filter woody plants only - Simplify taxonomy (one Genus, one species).





[1]:

```python
import pandas as pd
import matplotlib.pyplot as plt
from mpl_toolkits.basemap import Basemap

# comments
#this is the file path. data/ is the folder and TmpExpItem02.csv is the name of the file
file_path = 'data/TmpExpItem02.csv'
plants_data = pd.read_csv(file_path)

# Filter the dataset based on the 'LifeForm' column criteria
filtered_data = plants_data[plants_data['LifeForm'].isin(['Tree', 'Shrub', 'Shrub or Tree'])]

# Prepare the 'GenusSpecies' column for preprocessing
# Split the 'GenusSpecies' column into  (at least) separate words
split_genus_species = filtered_data['GenusSpecies'].str.split()

# Filter out records with not exactly 2 words
filtered_genus_species = split_genus_species[split_genus_species.apply(len) == 2]

# Rejoin the genus and species words into a single string seperated by a space " "
genus_species_final = filtered_genus_species.apply(lambda x: " ".join(x))
```

```python
# Create a new DataFrame with the cleaned 'GenusSpecies' column
final_df = genus_species_final.to_frame(name='GenusSpecies')

# Remove duplicate entries in 'GenusSpecies' to keep only unique values
unique_genus_species = final_df.drop_duplicates()

unique_genus_species.head()
```

[1]:

```
                GenusSpecies
0   Rhododendron racemosum
1    Rhododendron bureavii
2    Rhododendron fortunei
3      Rhododendron niveum
4  Rhododendron augustinii
```

[2]:

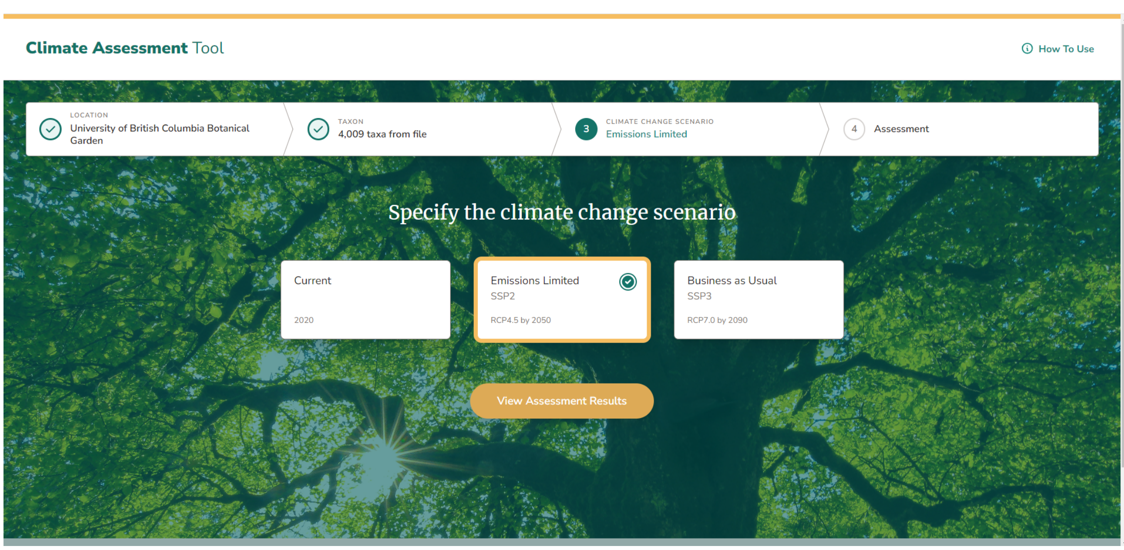
```python
#this cell downloads the preprocessed dataframe
"""# Define the file path for the exported CSV
export_file_path = 'data/preprocessed_genus_species.csv'

# Export the DataFrame to CSV
unique_genus_species.to_csv(export_file_path, index=False)
""";
```

1.2 Submit the Data to CAT

Choose location (name of institution or geo-location coordinates).

Input taxa - Website interface submit single taxon name or list of taxa in a CSV or Excel - If multiples the list is a Series (a file with a single column of data).

Choose a climate scenario (current, SPP2, SPP3).



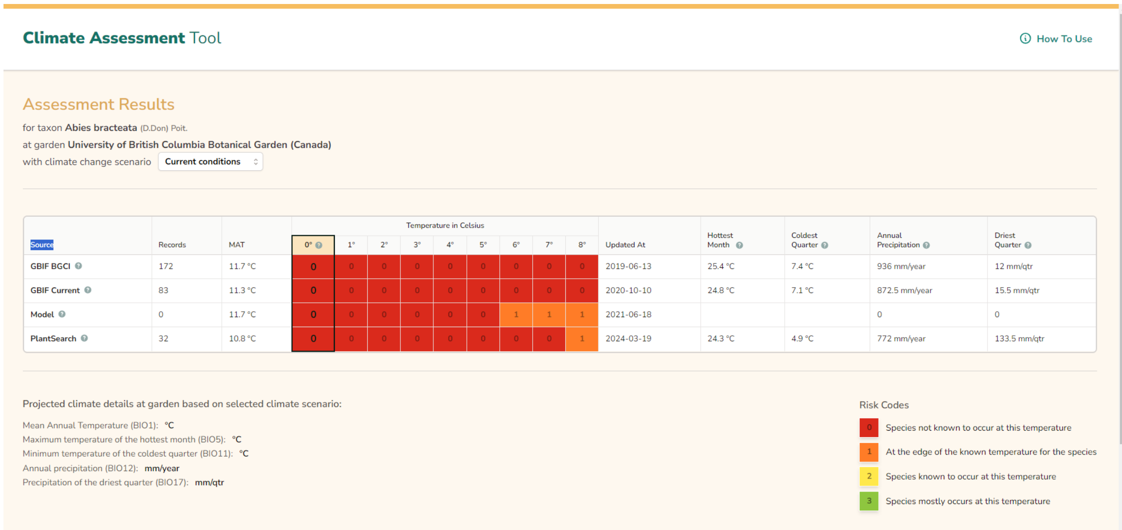
1.3 CAT Output

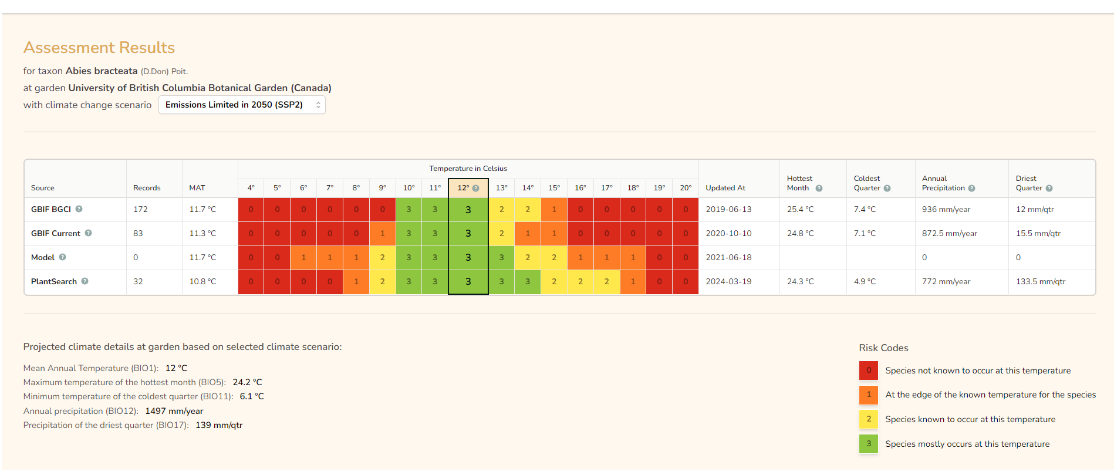
Single taxon output: - Infographic of the assessment.

Multiple taxon output: - Tabular representation - Output csv or excel file

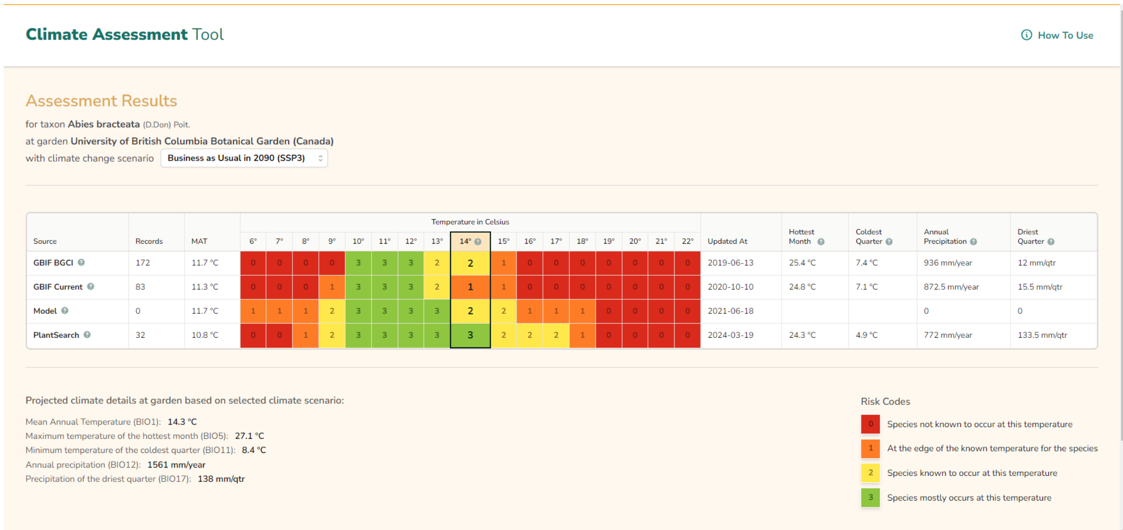
1.3.1 Single Taxon assesment:

Current condition infographic portion seems to infer that the species does not occur, this should be ignored. The assesement of a single taxon shows the number of observations used to calculate the mean, the source of the observations,the mean average temperature and several temperature and rain statistics.





SSP2



Climate Assessment Tool

How To Use

Assessment Results

for taxon **Abies bracteata** (D.Don) Poit.
at garden **University of British Columbia Botanical Garden (Canada)**
with climate change scenario **Business as Usual in 2090 (SSP3)**

Source	Records	MAT	Temperature in Celsius																	Updated At	Hottest Month	Coldest Quarter	Annual Precipitation	Driest Quarter
			6°	7°	8°	9°	10°	11°	12°	13°	14°	15°	16°	17°	18°	19°	20°	21°	22°					
GBIF BGCI	172	11.7 °C	0	0	0	0	3	3	3	2	2	1	0	0	0	0	0	0	0	2019-06-13	25.4 °C	7.4 °C	936 mm/year	12 mm/qtr
GBIF Current	83	11.3 °C	0	0	0	1	3	3	3	2	1	1	0	0	0	0	0	0	0	2020-10-10	24.8 °C	7.1 °C	872.5 mm/year	15.5 mm/qtr
Model	0	11.7 °C	1	1	1	2	3	3	3	3	2	2	1	1	1	0	0	0	0	2021-06-18			0	0
PlantSearch	32	10.8 °C	0	0	1	2	3	3	3	3	3	2	2	2	1	0	0	0	0	2024-03-19	24.3 °C	4.9 °C	772 mm/year	133.5 mm/qtr

Projected climate details at garden based on selected climate scenario:

Mean Annual Temperature (BIO1): **14.3 °C**
Maximum temperature of the hottest month (BIO5): **27.1 °C**
Minimum temperature of the coldest quarter (BIO11): **8.4 °C**
Annual precipitation (BIO12): **1561 mm/year**
Precipitation of the driest quarter (BIO17): **138 mm/qtr**

Risk Codes

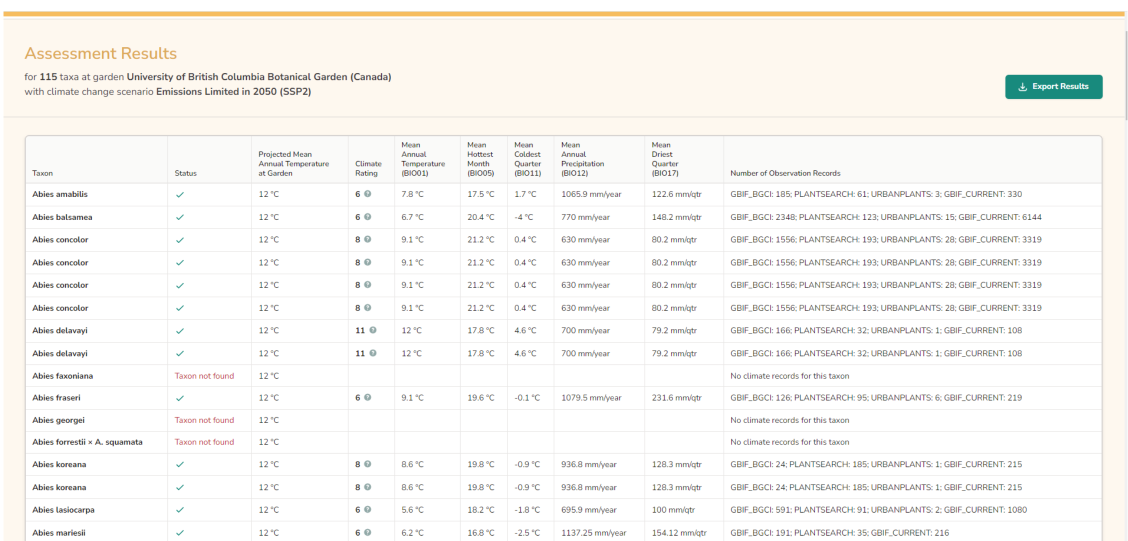
0 Species not known to occur at this temperature
1 At the edge of the known temperature for the species
2 Species known to occur at this temperature
3 Species mostly occurs at this temperature

SPP3

1.3.2 Multiple assesment

When subiting a list of taxa, CAT output is a tabulated form, with the option of exporting the data as a csv or excel file. In this format the observation is given a 'climate rating' which indicates the likelyhood of a given taxon to grow under the predicted mean anual temperature. The climate rating is a scale of how well the taxon will fit under a new climate regime, assuming that the taxon is represented by the environmental boundaries calculated (Mean Annual Temperature, Mean Hottest Month, Mean Coldest Quarter, Mean Annual Precipitation, Mean driest Month).

Similar problem to the single taxon output the 'current' model cannot be used as test for the compatability of taxa growing in the Garden right now. Until this 'bug' is addressed the 'climate rating' under the 'current conditions' model should not be used, and is not informative



Assessment Results

for **115** taxa at garden **University of British Columbia Botanical Garden (Canada)**
with climate change scenario **Emissions Limited in 2050 (SSP2)**

Export Results

Taxon	Status	Projected Mean Annual Temperature at Garden	Climate Rating	Mean Annual Temperature (BIO01)	Mean Hottest Month (BIO05)	Mean Coldest Quarter (BIO11)	Mean Annual Precipitation (BIO12)	Mean Driest Quarter (BIO17)	Number of Observation Records
Abies amabilis	✓	12 °C	6	7.8 °C	17.5 °C	1.7 °C	1065.9 mm/year	122.6 mm/qtr	GBIF_BGCI: 185; PLANTSEARCH: 61; URBANPLANTS: 3; GBIF_CURRENT: 330
Abies balsamea	✓	12 °C	6	6.7 °C	20.4 °C	-4 °C	770 mm/year	148.2 mm/qtr	GBIF_BGCI: 2348; PLANTSEARCH: 123; URBANPLANTS: 15; GBIF_CURRENT: 6144
Abies concolor	✓	12 °C	8	9.1 °C	21.2 °C	0.4 °C	630 mm/year	80.2 mm/qtr	GBIF_BGCI: 1556; PLANTSEARCH: 193; URBANPLANTS: 28; GBIF_CURRENT: 3319
Abies concolor	✓	12 °C	8	9.1 °C	21.2 °C	0.4 °C	630 mm/year	80.2 mm/qtr	GBIF_BGCI: 1556; PLANTSEARCH: 193; URBANPLANTS: 28; GBIF_CURRENT: 3319
Abies concolor	✓	12 °C	8	9.1 °C	21.2 °C	0.4 °C	630 mm/year	80.2 mm/qtr	GBIF_BGCI: 1556; PLANTSEARCH: 193; URBANPLANTS: 28; GBIF_CURRENT: 3319
Abies concolor	✓	12 °C	8	9.1 °C	21.2 °C	0.4 °C	630 mm/year	80.2 mm/qtr	GBIF_BGCI: 1556; PLANTSEARCH: 193; URBANPLANTS: 28; GBIF_CURRENT: 3319
Abies delavayi	✓	12 °C	11	12 °C	17.8 °C	4.6 °C	700 mm/year	79.2 mm/qtr	GBIF_BGCI: 166; PLANTSEARCH: 32; URBANPLANTS: 1; GBIF_CURRENT: 108
Abies delavayi	✓	12 °C	11	12 °C	17.8 °C	4.6 °C	700 mm/year	79.2 mm/qtr	GBIF_BGCI: 166; PLANTSEARCH: 32; URBANPLANTS: 1; GBIF_CURRENT: 108
Abies faxoniana	Taxon not found	12 °C							No climate records for this taxon
Abies fraseri	✓	12 °C	6	9.1 °C	19.6 °C	-0.1 °C	1079.5 mm/year	231.6 mm/qtr	GBIF_BGCI: 126; PLANTSEARCH: 95; URBANPLANTS: 6; GBIF_CURRENT: 219
Abies georgei	Taxon not found	12 °C							No climate records for this taxon
Abies forrestii × A. squamata	Taxon not found	12 °C							No climate records for this taxon
Abies koreana	✓	12 °C	8	8.6 °C	19.8 °C	-0.9 °C	936.8 mm/year	128.3 mm/qtr	GBIF_BGCI: 24; PLANTSEARCH: 185; URBANPLANTS: 1; GBIF_CURRENT: 215
Abies koreana	✓	12 °C	8	8.6 °C	19.8 °C	-0.9 °C	936.8 mm/year	128.3 mm/qtr	GBIF_BGCI: 24; PLANTSEARCH: 185; URBANPLANTS: 1; GBIF_CURRENT: 215
Abies lasiocarpa	✓	12 °C	6	5.6 °C	18.2 °C	-1.8 °C	695.9 mm/year	100 mm/qtr	GBIF_BGCI: 591; PLANTSEARCH: 91; URBANPLANTS: 2; GBIF_CURRENT: 1080
Abies mariesii	✓	12 °C	6	6.2 °C	16.8 °C	-2.5 °C	1137.25 mm/year	154.12 mm/qtr	GBIF_BGCI: 191; PLANTSEARCH: 35; GBIF_CURRENT: 216

1.3.3 Details of predicted climate (example 2050 reduced emmissions)

These details are specific to the geolocation chosen at the start of the process

Projected climate details at garden based on selected climate scenario:

Mean Annual Temperature (BIO1): **12 °C**

Maximum temperature of the hottest month (BIO5): **24.2 °C**

Minimum temperature of the coldest quarter (BIO11): **6.1 °C**

Annual precipitation (BIO12): **1497 mm/year**

Precipitation of the driest quarter (BIO17): **139 mm/qtr**

1.3.4 Climate Rating Scale

Note that the range starts with the worst case 0 (not likely) to best case 11 (middle of natural range)

Climate Rating

0	Not known and not likely	**6**	Shoulder of BG range
1	Not known but possible	**7**	Shoulder of urban range
2	Not known but likely	**8**	Shoulder of natural range
3	Near edge of BG range	**9**	Middle of BG range
4	Near edge of urban range	**10**	Middle of urban range
5	Near edge of natural range	**11**	Middle of natural range

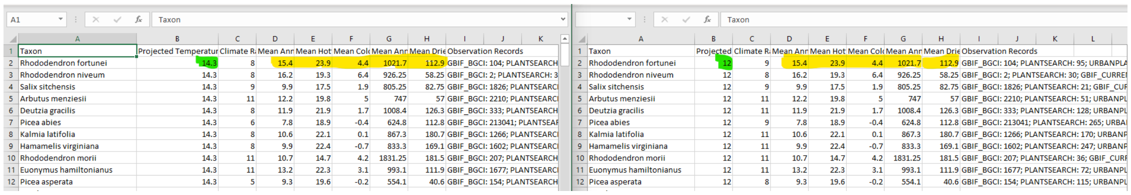
1.4 Feature Engineering

Concatenation: - For a single CAT report. - For multiple CAT reports.

Below are example CAT output of both SPP3 (left) and SPP2 (right) climate scenarios. Note that the green highlight is the Mean Average Temperature from the model, which will be different in both outputs (as is the predicted climate rating). Also not that the yellow highlited features in both data sets are identical. These represent the niche space in which the taxon is thought best to occur.

Since the projected temperature is the same for the entire dataset (model dependent), and the all of the other features are the same between models, with the exceptionm of the Climate Rating, concatinating is relaively simple. In excel you would simply cut and paste one of the Climate

Rating columns from one of the data sets into the other (ensuring your data was properly lined-up before hand), change the 'Climate rating' feature name to reflect the model of origin and drop the 'projected temperature' feature to avoid confusion.



[3]:
```python
# Load the sample datasets
ssp2_data = pd.read_csv('data/sampled_genus_species_results_SSP2.csv')
ssp3_data = pd.read_csv('data/sampled_genus_species_results_SSP3.csv')

# Rename 'Climate Rating' to 'rating_SSP2' and 'rating_SSP3' respectively
ssp2_data.rename(columns={'Climate Rating': 'rating_SSP2'}, inplace=True)
ssp3_data.rename(columns={'Climate Rating': 'rating_SSP3'}, inplace=True)

# Select Taxon and rating_SSP3 from SSP3
ssp3_rating = ssp3_data[['Taxon', 'rating_SSP3']].copy()

# Merge SSP3 rating into SSP2 dataset on Taxon
merged_data = pd.merge(ssp2_data, ssp3_rating, on='Taxon', how='left')

# Drop unwanted column
merged_data = merged_data.drop(columns="Projected Temperature, °C")

# Reorder columns to place 'rating_SSP3' next to 'rating_SSP2'
columns = list(merged_data.columns)
rating_ssp2_index = columns.index('rating_SSP2')
# Insert 'rating_SSP3' right after 'rating_SSP2'
columns.insert(rating_ssp2_index + 1, columns.pop(columns.index('rating_SSP3')))
# Reassign the ordered columns to the DataFrame
merged_data = merged_data[columns]

merged_data
```

[3]:
```
                       Taxon  rating_SSP2  rating_SSP3  \
0           Cotinus obovatus            9           11
1     Chamaecyparis thyoides           11           11
2              Acer oblongum            9            9
3    Rhododendron sinogrande           11           11
4              Acer palmatum           11           11
5          Pinus wallichiana            9            6
6           Pinus sylvestris            6            6
7        Aesculus californica            9           11
8      Photinia beauverdiana            9           11
```

9	Tamarix ramosissima	9	11
10	Pinus resinosa	6	3

	Mean Annual Temperature (BIO01), °C	Mean Hottest Month (BIO05), °C \
0	13.3	24.8
1	11.7	23.4
2	15.9	23.6
3	11.7	16.5
4	12.6	22.6
5	11.6	25.5
6	8.2	19.0
7	14.2	21.9
8	15.5	23.3
9	12.3	24.0
10	7.3	21.4

	Mean Coldest Quarter (BIO11), °C \
0	2.0
1	1.1
2	6.0
3	3.9
4	2.7
5	4.4
6	NaN
7	7.0
8	4.9
9	2.5
10	-3.7

	Mean Annual Precipitation (BIO12), mm/year \
0	831.30
1	913.60
2	1154.30
3	791.50
4	965.30
5	874.75
6	601.60
7	548.50
8	1053.60
9	363.40
10	747.00

	Mean Driest Quarter (BIO17), mm/qtr \
0	156.30
1	194.80
2	94.80
3	72.62

```
4                                130.10
5                                145.25
6                                109.60
7                                 33.00
8                                109.20
9                                 39.50
10                               139.20

                                  Observation Records
0   GBIF_BGCI: 47; PLANTSEARCH: 128; URBANPLANTS: …
1   GBIF_BGCI: 338; PLANTSEARCH: 115; URBANPLANTS:…
2   GBIF_BGCI: 225; PLANTSEARCH: 49; URBANPLANTS: …
3   GBIF_BGCI: 106; PLANTSEARCH: 42; GBIF_CURRENT: 48
4   GBIF_BGCI: 1368; PLANTSEARCH: 266; URBANPLANTS…
5                    PLANTSEARCH: 169; URBANPLANTS: 8
6   GBIF_BGCI: 272728; PLANTSEARCH: 237; URBANPLAN…
7   GBIF_BGCI: 2060; PLANTSEARCH: 84; URBANPLANTS:…
8   GBIF_BGCI: 696; PLANTSEARCH: 45; URBANPLANTS: …
9   GBIF_BGCI: 115; PLANTSEARCH: 44; URBANPLANTS: …
10  GBIF_BGCI: 789; PLANTSEARCH: 115; URBANPLANTS:…
```

1.4.1 Re-annealing with original dataset

To be able to reconnect the CAT output with an irisBG report that produced it (and recover the connection between taxa and accession number) we need to be aware of a few things - The simplified nomenclature used to produce the CAT report may no longer be reflective of that taxon. this is particularly true of taxon that have had subsp. removed during preprocessing. - Taxon and accession number are not necessarily one-to-one. A single taxon may have several accession numbers, and several item numbers with each accession. If the goal is to relay the information provided by CAT, ie. the 'Climate rating' back to accessioned items in irisBG then care needs to be taken to make sure each instance of a taxon in the inventory be considered.

[4]:
```python
#load files
irisBG_report_df = pd.read_csv('data/TmpExpItem02.csv')
df2 = pd.read_csv('data/preprocessed_genus_species_SPP2.csv')

# Renaming 'Taxon' to 'GenusSpecies' in the CAT CSV
df2_renamed = df2.rename(columns={'Taxon': 'GenusSpecies'})

# Merging 'Climate Rating' from the second CSV to the first CSV based on␣
↪'GenusSpecies'
merged_df = pd.merge(irisBG_report_df, df2_renamed[['GenusSpecies', 'Climate␣
↪Rating']], on='GenusSpecies', how='left')

merged_df
```

```
[4]:        AccNoFull              GenusSpecies ItemLocationCode  LocationCoordY  \
     0      1965-0043   Rhododendron racemosum             LAS5     -123.246133
     1      1966-0043    Rhododendron bureavii             3AD4     -123.248276
     2      1969-0009    Rhododendron fortunei             3AH4     -123.248846
     3      1970-0005      Rhododendron niveum             3AA4     -123.249978
     4      1972-0052  Rhododendron augustinii             3AB5     -123.249410
     …            …                        …              …               …
     13995  2012-0580         Acer pentaphyllum                8     -123.234415
     13996  2024-0121          Fargesia robusta             3AAG     -123.250404
     13997  1985-0433         Camellia japonica             3AC3     -123.250045
     13998  2002-0466        Paeonia jishanensis             3AC4     -123.250129
     13999  2021-0026         Lomatium dissectum             1G02     -123.247253

            LocationCoordX             LifeForm  Climate Rating
     0           49.253477                Shrub             NaN
     1           49.252149                Shrub             NaN
     2           49.251486        Shrub or Tree             9.0
     3           49.253652                 Tree             8.0
     4           49.253041                Shrub             NaN
     …                 …                    …              …
     13995       49.249515        Shrub or Tree             9.0
     13996       49.253779  Herbaceous Perennial            NaN
     13997       49.252760        Shrub or Tree            11.0
     13998       49.252658                Shrub             NaN
     13999       49.252857  Herbaceous Perennial            NaN

     [14000 rows x 7 columns]
```

Note There are 9,113 NaN entries in the 'Climate Rating' column of the merged dataframe. These represent the herbaceous accessions, accessions with missing lifeform observation and taxa not included in CAT.

```
[5]: # Dropping rows with NaN values in the 'Climate Rating' column
     merged_df_cleaned = merged_df.dropna(subset=['Climate Rating'])

     # Displaying the number of rows after dropping NaN values in 'Climate Rating'
     merged_df_cleaned.shape
```

```
[5]: (4887, 7)
```

```
[6]: # Counting the number of records in each of the 11 climate rating scores
     climate_rating_counts = merged_df_cleaned['Climate Rating'].value_counts().
      ↪sort_index()

     climate_rating_counts
```

[6]:
```
Climate Rating
2.0       13
3.0       48
5.0       52
6.0      524
8.0      327
9.0     1562
11.0    2361
Name: count, dtype: int64
```

1.4.2 Climate rating 3 or below

In CAT a climate 3 or below is either at the edge of the range, or unlikely altogether

[7]:
```python
# Creating a new dataframe with observations having climate ratings of 3 and␣
↪below
unlikely_df = merged_df_cleaned[merged_df_cleaned['Climate Rating'] <= 3]

# Displaying the number of rows in unlikely_df to confirm
unlikely_df
```

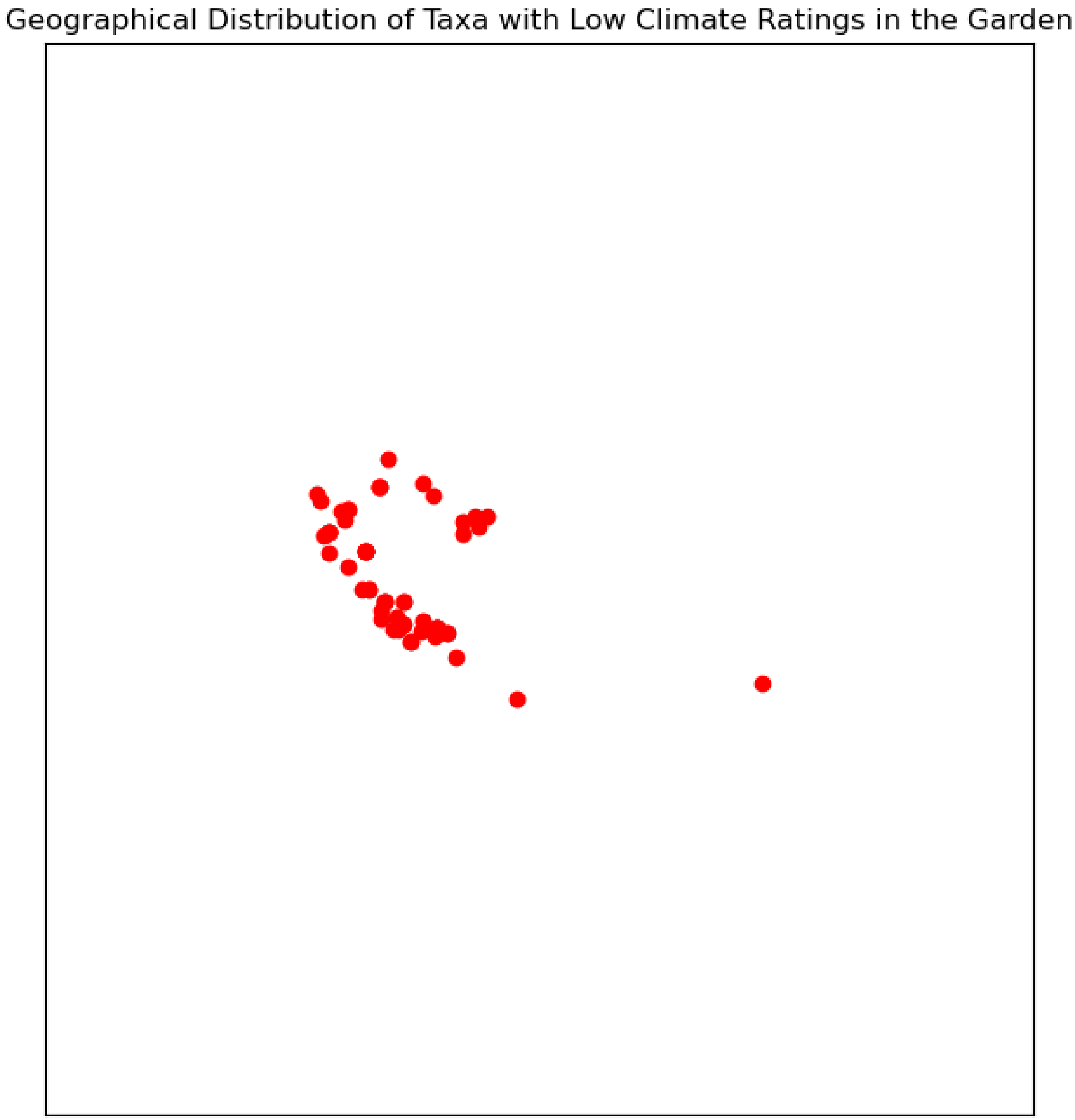
[7]:
```
         AccNoFull             GenusSpecies ItemLocationCode  LocationCoordY  \
215    2000-0476  Rhododendron sikangense             3AC7     -123.249040
493    1965-0004       Rhododendron erosum             3AB6     -123.250395
538    1972-0208    Rhododendron hodgsonii             3AC7     -123.249040
853    1990-0574          Tilia chingiana             3AF7     -123.248057
1112   2000-0625  Rhododendron sikangense             3AE8     -123.248383
…            …                        …                …               …
11099  1965-0004       Rhododendron erosum             3AB6     -123.250395
11964  2021-0625          Dypsis decipiens               2A     -123.250883
12473  2022-0160       Crataegus submollis                8     -123.234415
13335  2023-0097     Rhododendron spilotum             3AK4     -123.245750
13386  2023-0097     Rhododendron spilotum             3AB8     -123.250583

       LocationCoordX       LifeForm  Climate Rating
215         49.252716          Shrub             3.0
493         49.253165          Shrub             3.0
538         49.252716          Shrub             3.0
853         49.250835           Tree             3.0
1112        49.251504          Shrub             3.0
…                 …             …             …
11099       49.253165          Shrub             3.0
11964       49.254102           Tree             2.0
12473       49.249515  Shrub or Tree             3.0
13335       49.250164           Tree             3.0
13386       49.253088           Tree             3.0
```

```
[61 rows x 7 columns]
```

1.5 Data Visualization

Examples: - Heat maps created by Ben S. last fall included all of the Climate Ratings (which can be deceiving). Climate rating is an ordinal score, but really we are only concerned with 3 and below (these represent taxa that will likely have a harder time growing under warming conditions). - Mapping specific individual might be more informative - example below (without a reference map)

[8]:
```
import matplotlib.pyplot as plt
from mpl_toolkits.basemap import Basemap

# Setting up the plot
plt.figure(figsize=(10, 8))

# Creating a Basemap instance for plotting
# Since the locations are within the UBCBG, we'll use a small area for␣
↪demonstration purposes
# Adjust lat_0 and lon_0 to the garden's approximate center, if known
lat_0, lon_0 = unlikely_df['LocationCoordX'].mean(),␣
↪unlikely_df['LocationCoordY'].mean()
m = Basemap(projection='merc', llcrnrlat=unlikely_df['LocationCoordX'].min() -␣
↪0.01, urcrnrlat=unlikely_df['LocationCoordX'].max() + 0.01,
            llcrnrlon=unlikely_df['LocationCoordY'].min() - 0.01,␣
↪urcrnrlon=unlikely_df['LocationCoordY'].max() + 0.01, lat_ts=20,␣
↪resolution='c')

m.drawcoastlines()
m.drawcountries()
m.drawstates()

# Converting latitude and longitude to x and y for plotting
x, y = m(unlikely_df['LocationCoordY'].values, unlikely_df['LocationCoordX'].
↪values)

# Plotting the points
m.scatter(x, y, marker='o', color='red', zorder=5)

plt.title('Geographical Distribution of Taxa with Low Climate Ratings in the␣
↪Garden')
plt.show()
```



1.6 Concerns

1.6.1 Domain Problems:

- Ignores local climatic zones within a garden.
- Does not account for outliers, such as average temperature and average rainfall.
- Makes generalized predictions without considering local fluctuations.
- Shows bias towards more observations from sites in the Northern Hemisphere.
- Records may include plants under protection, potentially skewing results.

1.6.2 CAT Search Problems:

Nomenclature:

- Species name is spelled incorrectly.
- The species name contains a special character.
- The species name is used in a synonym form.
- An infraspecific name beyond the species-level binomial has b

Type:

- The species is not classified as a 'tree'.

Climate Change Alliance of Botanic Gardens. [2024]. Climate Assessment Tool v1.Botanic Gardens Conservation International. Richmond, U.K. Available at https://cat.bgci.org.Accessed on 20/03/2024.

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